Energy-Storage Turbine System: Calculations

Properties of Car and Turbine

Properties of Car	Properties of Turbine
Mass of car = 1500 kg	Diameter = 0.3 m
Top speed = 330 km/h ≈ 91.67 m/s	Mass per blade = 0.045 kg
	Mass of fan = 0.2 kg

Mass of System

- Rod/Shaft = 0.5 kg
- Mass of fan = 0.2 kg
- Pinion = 0.08 kg
- Bevel gear = 0.075 kg
- Rack = 1 kg
- **Whole system** = 0.5 + 0.2 + 0.08 + 0.075 + 1 = **1.855 kg**

Average Pressure Exerted on the Turbine

$P = \rho v^2/2$, where ρ = air density = 1.225 kg/m³ and v = average velocity = (91.67 + 0)/2 = 45.84 m/s

$P = 1.225\ (45.84)^2 / 2 =$ **1287 Pa**

Force Exerted on Turbine

$P = F/A$, where $A = \pi r^2 = \pi(0.15)^2 = 9\pi/400$

$1287 = F / (9\pi/400)$, so **F = 91 N**

Acceleration of the System

$F = ma \rightarrow 91 = 1.855a \rightarrow$ **a = 49 m/s²**

Linear Velocity Square (v^2) of Turbine

$a = v\omega$ and $v = r\omega$, so $\omega = a/v$, $v = r(a/v)$, hence $v^2 = ra$

Midpoint (average) of turbine blade radius = (0 + 0.15)/2 = 0.075 m. We did this to find the average linear velocity of the turbine, so we took the median radius.

$v^2 = (0.075)(49) =$ **$3.675\ m^2/s^2$**

$(x - 3.675)/3.675 \times 100 = 33 \rightarrow$ **$x = 4.89\ m^2/s^2$**

∴ Calculus states that when calculating the v^2 value using the median radius, the actual value is 33% larger than the calculated value, but the percentage difference is constant for all values of v^2.

Angular Velocity of Turbine

$\omega = v/r \rightarrow \omega^2 = v^2/r^2 \rightarrow \omega^2 = 4.89 / (0.075)^2 \rightarrow$ **$\omega = 29.48$ rad/s**

Revolutions per second = 29.48 / 2π = **4.692 rev/s**

Forward Thrust

$T = k_T \rho n^2 D^4$, where k_T = thrust coefficient = 0.6, ρ = air density = 1.225 kg/m^3, n = revolutions per second = 4.692

$T = (0.6)(1.225)(4.692)^2(0.3)^4 =$ **0.131 N**

Drag Force Caused by Fan

D.F. = $\rho \times v^2 \times C_D \times A / 2$, where C_D = drag coefficient = 0.05 and $A = \pi r^2 = 9\pi/400$

D.F. = $(1.225)(4.89)^2(0.05)(9\pi/400) / 2 =$ **0.0518 N**

Net Acceleration

$F = ma \rightarrow 0.131 - 0.0518 = 1500a \rightarrow 0.0792 = 1500a \rightarrow$ **$a = 5.28 \times 10^{-5}\ m/s^2$**

Notes

- For the fan-created forward thrust, I used the thrust and drag coefficients of fans on drones.

- **Material:** Using plastic was expensive, hence we used steel. However, plastic is lightweight, so it would make the system more effective. Plastic can also deteriorate faster than steel.